

Project 2

CS 261

Save your program for this project as `<emailID>_project2.py` where `<emailID>` is the part of your Hampden-Sydney e-mail address before the `@` symbol. When you are finished, e-mail your program to `blins@hsc.edu`. Your solution is due by 3:00pm on Friday, October 2.

Caesar Cipher

The Caesar cipher is one of the oldest and simplest methods for encrypting text. You pick a shift amount and then you shift every letter in the message by that amount. For example, if the shift is 2, then a's become c's, b's become d's and so on, until y and z wrap around to become a and b, respectively. A message like:

it's about to rain

would become

kv'u cdqvw vq tckp

For this project, you will write four functions related to the Caesar cipher. You may also include any additional functions that are helpful to construct the functions described below. You should not import any libraries for this project.

1. Write a function `letter2number` that converts lower case letters (a, b, c, ..., z) to the numbers (0, 1, 2, ..., 25). For example `letter2number("c")` should return 2.
2. Write a function `number2letter` that converts a number to the corresponding (lower case) letter of the alphabet. For example, `number2letter(1)` should return "b".
3. Write a function `caesar_shift(message, shift)` that applies a Caesar shift to any string. You can assume that the message will not have any upper case letters, and you only need to shift letters, not any other symbols.
4. The Caesar cipher is not a good way to encrypt a message because it is very easy to decrypt. Write a function called `try_all_shifts(message)` that will apply every Caesar shift from 1 to 25 to a string and print out the results. Use your function to decrypt the following multiline string (which you can copy into your program).

```
mystery = """gurer ner fgenatr guvatf qbar va gur zvqavtug fha
ol gur zra jub zbvy sbe tbyq;
gur nepgvp genvyf unir gurve frperg gnyrf
gung jbhyy znxr lbhe oybbq eha pbyq;
gur abegurea yvtugf unir frra dhrre fvtugf,
ohg gur dhrrerfg gurl rire qvq frf
jnf gung avtug ba gur znetr bs ynxr yronetr
v perzngrq fnz zprrr."""
```

Here is a link with some tips on how to get started:

<https://bclins.github.io/fall26/coms261/project2tips.pdf>